

Ultimate Technologies Inc. Take-Home Challenge

Part 2 - Experiment and metrics design - answers by Steven Goldsmith

1) What would you choose as the key measure of success of this experiment in encouraging driver partners to serve both cities, and why would you choose this metric?

The key measure of success would be to increase the percentage of active driver partners who complete at least one trip in both cities within a given time period (for example weekly). The goal is to encourage drivers to be available in both cities, and this metric captures that. By taking the percentage of multiple drivers instead of just the total count increase of trips across the bridge, we avoid the experiment being dominated by only a few users doing many trips.

2) Describe a practical experiment you would design to compare the effectiveness of the proposed change in relation to the key measure of success. Please provide details on:

a) how you will implement the experiment

We would use a randomized control trial at the driver level. Start with the population of drivers who operate primarily in 1 city (exclude drivers who already regularly operate in both cities as there is less likelihood of demonstrable behavior change). Then stratify the drivers by home city and typical driving patterns. Randomly split the drivers into two groups: Control group, where there is no toll reimbursement; Treatment group, where all tolls are reimbursed for bridge crossings for a specified period (for example 2 months). Prior to starting the experiment, collect baseline data for 1-2 months and notify treatment group drivers of the reimbursement of tolls.

b) what statistical test(s) you will conduct to verify the significance of the observation

We would conduct 2 statistical tests on the data.

1. Two proportion z-test to compare the proportion of drivers operating in both cities between the control group and treatment group.
2. Difference in differences test to compare the change (pre/during the experiment) in the control group to the change in the treatment group.

c) how you would interpret the results and provide recommendations to the city operations team along with any caveats.

The results of the experiment would be evaluated by determining whether toll reimbursement led to a statistically significant increase in the percentage of driver partners serving both Gotham and Metropolis compared to the control group. If the treatment group shows a meaningful improvement, this would indicate that toll costs were a significant barrier preventing drivers from operating across both cities. In this case, Ultimate should consider expanding the program in a phased rollout to validate at scale, while analyzing whether the increased driver availability improves rider wait times, trip completion rates, and revenue enough to justify the reimbursement expense.

If the program increases cross-city activity but the cost of reimbursement outweighs the business benefit, Ultimate could consider a more targeted approach, such as offering

reimbursements only during peak demand periods or for specific trips where additional coverage is most valuable.

If the experiment does not produce a significant increase in cross-city driver activity, it would suggest that toll costs are not the primary factor limiting driver movement between cities. Ultimate should then explore other solutions, such as targeted incentives, improved communication about demand opportunities, or additional driver benefits.

The results should also be interpreted with consideration for potential limitations. Driver behavior may temporarily change during the experiment but revert afterward (novelty effect), so a post-experiment analysis would help determine whether the impact is sustainable. Additionally, external factors such as seasonal demand, holidays, or events could influence driver behavior. Segmenting results by weekday/weekend and time of day would help determine when toll reimbursement provides the greatest value.